



Birla Institute of Technology & Science, Pilani

Pilani Campus

Academic Undergraduate Studies Division

SECOND SEMESTER 2026-2027

COURSE HANDOUT (PART-II)

Date: 1 Aug 2026

In addition to the Part I (The general handout for all courses appended to the timetable), this portion gives further details regarding the course.

Course No. : PHY F318

Course Title : Atoms and Photons

Instructor In-charge : Amol R Holkundkar

AIMS AND LEARNING OBJECTIVE:

Quantum mechanical understanding of light in terms of photons, sources of light from atomic emissions, two-level systems, parametric down-conversion, one-photon and two-photon processes, quantum measurements, photon detection, counting and correlation measurements, photon statistics, single photon interference, phase and polarization, entanglement and EPR experiment, Bell's theorem, quantum cryptography and teleportation, atomic clock, quantum imaging.

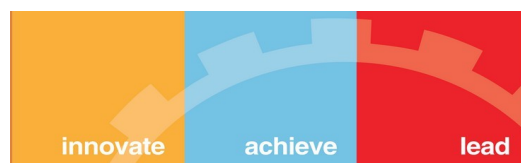
Textbook: Quantum Optics, Marlan O. Scully and M. Suhail Zubairy

Reference Books:

- The Quantum Theory of Light, Rodney Loudon, Third Edition, Oxford University Press, 2010.
- Quantum Optics, Miguel Orszag, Third Edition, Springer, 2016.
- Atom-Photon interactions, Claude Cohen-Tannoudji, Wiley-VCH.

Course Plan:

Lecture No.	Learning Objectives	Topics to be Covered	Learning Outcome	Reference to Text and Reference Book
1-6	Field Quantization	Canonical quantization of the e-m fields, the single and multi mode field, Quantum fluctuation, Thermal fields, The	To learn the quantization of the electromagn	TB : Ch. 1.





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		quantum phase.	etic field to get the picture of photons.	
7-12	Atom-Field Interactions	Interaction of an atom with a classical field, interaction of an with a quantized field, The Rabi model, The Jaynes-Cummins model, Extension of the Jaynes-Cummins model.	To understand the interaction of light with atoms both in the semiclassical as well as fully quantum mechanical approach.	TB : Ch. 5, 6
13 – 18	Coherent states and Squeezed Coherent states	Coherent states of an oscillator, phase space picture of coherent states, Coherent states of light, Generation of coherent states.	To understand the important concept of coherent states which forms the basis of the squeezed states.	TB : Ch. 2
19 – 22	Quantum Theory of Coherence	Classical and quantum coherence functions, Young's interference, Higher order coherence functions.	To understand the important concept coherence and noncoherence of light both in classical and quantum theory.	TB : Ch. 2-4
23 - 30	Beam splitters and interferometers	Quantum mechanics of beam splitters, interferometry with a single photon, Interferometry with coherent states of light, Mach-Zehnder and Brown-Twiss interferometers.	To understand the quantum theory of interferometers and beam splitters.	TB : Ch 21
30 - 40	Optical tests of quantum mechanics	Photon sources, The quantum eraser, Induced coherence, Optical test of Bell's theorem.	To study the recent optical tests of quantum mechanics.	TB: Ch 20



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Evaluation Scheme:

Evaluation Component	Duration	Weightage	Date & Time	Nature of Test
Mid-Semester Test	90 Minutes	30%	10 th Oct 2025	Closed Book
Comprehensive Examination	3 Hours	40%	13 th Dec 2025	Partly closed and partly open
Tutorial Tests	Tutorial Tests 30 minutes each	30%	TBA a week prior	Closed Book Tutorial Tests

NC Policy: Minimum 15 % of the Course Total including attendance in 50 % of Tutorial Tests, Midterm, and Comprehensive examination.

Chamber Consultation Hour: Feel free to drop a mail to seek prior appointment for meeting.

Notices: Notices and solutions will be displayed only on **Google Classroom** only. All the necessary class notes will be uploaded on the Google Classroom after the class.

Instructor in Charge

PHY F318

